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$z^2 - 5z + 8 = -2i(z+5) \Leftrightarrow z^2 + (-5+2i)z + 8 + 10i = 0$

$D = (-5+2i)^2 - 4 \cdot (8+10i) = -11 - 60i$

$\Rightarrow \lambda^2 + 11\lambda - 900 = 0$

$\lambda_1 = \dfrac{-11+61}{2} = 25 \Rightarrow x = \pm 5$

$\lambda_2 = \dfrac{-11-61}{2} = -36 \Rightarrow y = \pm 6$

$\Rightarrow -5 + 6i$ en $5 - 6i$

$z_1 = \dfrac{5 - 2i - 5 + 6i}{2} = 2i$

$z_2 = \dfrac{5 - 2i + 5 - 6i}{2} = 5 - 4i$

References